Ride-Height Control of a Fully Submerged Hydrofoil Boat: LQR with Integral Action and a Kalman Filter

ELEC3304 Control — Assessment 2
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This document continues *model_derivation.pdf*, which derives the nonlinear model, the trim point and the LTI model $\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\tilde{\boldsymbol{u}} + E\boldsymbol{w}$, $\boldsymbol{y} = C\boldsymbol{x}$ with $\boldsymbol{x} = [\,h \;\; v \;\; \theta \;\; q\,]^\top$, two foil inputs and three measurements (bow height, pitch angle, pitch rate). Section 6 records the design decisions that changed during the work and the trade-offs behind them.

1 Objectives and specifications

The controller must hold the boat at a commanded ride height with the hull level, using only the three sensors, in the presence of waves and of errors in the model. We translate this into the following specifications, checked on the nonlinear model with actuator limits:

S1. **Stability**: the open-loop plant has an unstable pole; all closed-loop poles must lie in the left half-plane, also for speeds 6–9 m/s and masses 190–240 kg.

S2. **Tracking**: a 5 cm height step settles (2 %) within about 2 s with less than 10 % overshoot.

S3. **Zero steady-state error** in height and pitch when the speed or the mass differs from the design values.

S4. **Actuator limits**: in normal operation the foil deflection stays inside the 3.5° upward margin available at trim, and the rate inside 20 °/s.

S5. **Disturbance rejection**: in regular head waves (5 cm amplitude) the height error stays well below the 15 cm range in which the linear model is valid.

2 Controller

2.1 Structure

The controller is an LQR state feedback with integral action on two outputs, a reference feedforward and a reference prefilter. All quantities are deviations from trim:

$$\tilde{\boldsymbol{u}} = \boldsymbol{N}_u r_f - K\big(\hat{\boldsymbol{x}} - \boldsymbol{N}_x r_f\big) - K_i\,\boldsymbol{\xi}, \tag{1}$$

$$\dot{\boldsymbol{\xi}} = \begin{bmatrix} \hat{h} - r_f \\ y_2 \end{bmatrix}, \qquad \dot{r}_f = \frac{r - r_f}{t_f}. \tag{2}$$

Here r is the commanded height change, r_f its filtered version, $\hat{\boldsymbol{x}}$ the estimate from the observer of Section 3, and y_2 the measured pitch angle. The terms have the following roles:

- $K(\hat{\boldsymbol{x}} - \boldsymbol{N}_x r_f)$ stabilises the boat and drives the state to its target.

- $K_i\boldsymbol{\xi}$ removes steady-state errors in height and pitch caused by model errors (specification S3). Two integrators are needed because there are two outputs to hold; Section 6 explains why the height integrator acts on the estimate $\hat{h}$ and not on the raw ultrasonic reading.
- $\boldsymbol{N}_x$, $\boldsymbol{N}_u$ give the steady state that corresponds to a height r with level pitch, obtained from $\begin{bmatrix} A & B \\ C_z & 0 \end{bmatrix}\begin{bmatrix} \boldsymbol{N}_x \\ \boldsymbol{N}_u \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ [1\ \ 0]^\top \end{bmatrix}$ with $C_z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. They let the controller move directly to the new height instead of waiting for the integrators.
- The prefilter with $t_f = 0.3\,\mathrm{s}$ turns a step command into a smooth one, which keeps the foil deflection small (Section 6).
- **Anti-windup**: whenever the commanded foil angle exceeds the position limit, $\dot{\boldsymbol{\xi}}$ is set to zero (integrator clamping).

2.2 LQR design

The gains minimise $J = \int_0^\infty \left(\boldsymbol{x}_a^\top Q\, \boldsymbol{x}_a + \tilde{\boldsymbol{u}}^\top R\, \tilde{\boldsymbol{u}}\right) dt$ for the augmented model $\boldsymbol{x}_a = [\,\boldsymbol{x}; \boldsymbol{\xi}\,]$,

$$A_a = \begin{bmatrix} A & 0 \\ C_z & 0 \end{bmatrix}, \qquad B_a = \begin{bmatrix} B \\ 0 \end{bmatrix},$$

which is controllable (rank 6). The weights follow Bryson's rule, each diagonal entry being the inverse square of the largest acceptable value (Table 1); the two integral weights were then increased by a factor of 10 to speed up the removal of steady-state errors.

Table 1: LQR weights (Bryson's rule).

Weighted quantity	Acceptable value	Weight	Note
heave h	2 cm	$1/0.02^2$	
heave rate v	0.5 m/s	$1/0.5^2$	
pitch θ	2°	$1/(2°)^2$	
pitch rate q	10 °/s	$1/(10°)^2$	
$\int(h - r)$	—	$10/0.02^2$	tuned
$\int \theta$	—	$10/(1°)^2$	tuned
each foil deflection	3°	$1/(3°)^2$	margin is 3.5°

The resulting gains (angles in rad, lengths in m) are

$$K = \begin{bmatrix} 3.284 & 0.188 & 2.788 & 0.183 \\ 1.648 & 0.098 & -2.788 & -0.212 \end{bmatrix}, \qquad K_i = \begin{bmatrix} 7.443 & 4.155 \\ 3.626 & -8.529 \end{bmatrix},$$

and the closed-loop poles of the augmented system are -3.32, $-3.90 \pm 2.35\mathrm{j}$, $-10.83 \pm 5.60\mathrm{j}$ and -18.50. The gain structure is physical: a height error is corrected by both foils in the same direction (first column of K positive in both rows), while a pitch error is corrected by moving the foils in opposite directions (third column of opposite sign).

2.3 Results with full-state feedback

Before adding the observer the controller was tested with the true state, to separate the two designs.

Step response (Figure 1). A 5 cm step settles in 0.72 s with 1.9 % overshoot on the linear model; the nonlinear model with actuator limits is almost identical and the peak foil deflections are 1.5° (front) and 1.1° (rear), well inside the margin. S2 and S4 are met.

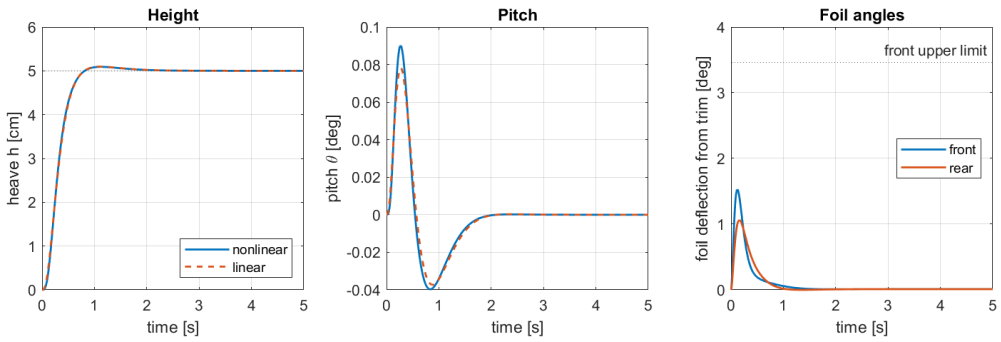


Figure 1: 5 cm height step with full-state LQR: linear and nonlinear model.

Waves (Figure 2). In regular head waves the open-loop boat drifts away, whereas the controlled boat holds the height within 0.18 cm rms and the pitch within 0.20° rms, with foil deflections up to 1.3°.

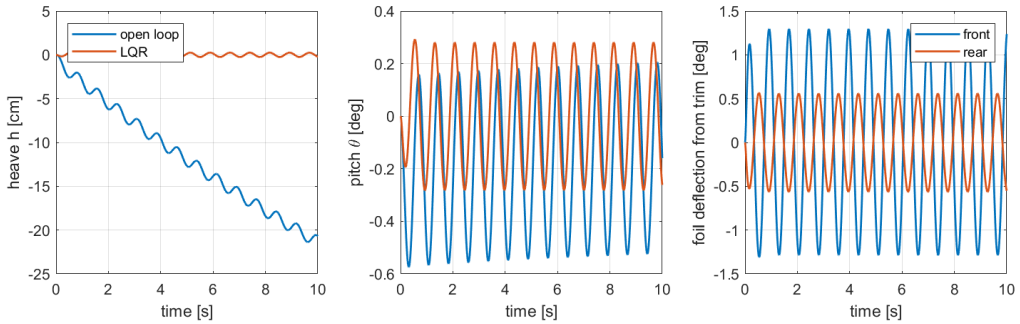


Figure 2: Regular head waves: open loop and full-state LQR.

Robustness (Figure 3). With the nominal gains, the closed loop of the true linearised plant is stable over the whole grid $U = 5.9$–$9\,\mathrm{m/s}$, $m = 190$–$240\,\mathrm{kg}$; the slowest pole is -3.17. In nonlinear simulation, a speed of 6 or 8 m/s causes an initial disturbance of at most 1.5 cm, which the integrators remove within about a second.

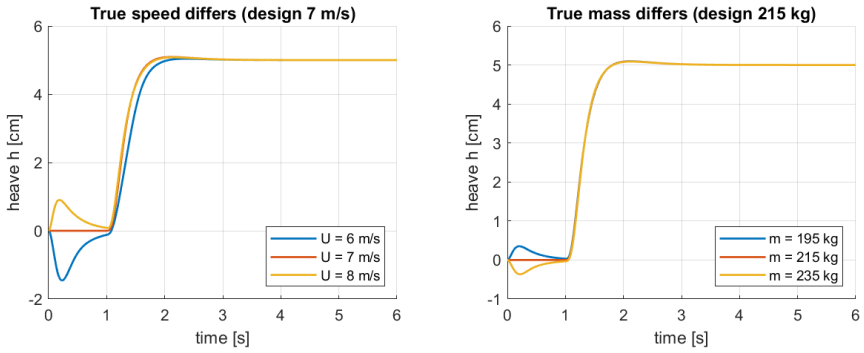


Figure 3: 5 cm step with speed and mass different from the design values (full-state LQR, trim inputs for the nominal values).

Saturation (Figure 4). A 20 cm step without prefilter drives the front foil to its 7° limit. Without anti-windup the integrators keep growing during saturation and the height overshoots by 8.7 cm; with clamping the overshoot is 1.0 cm.

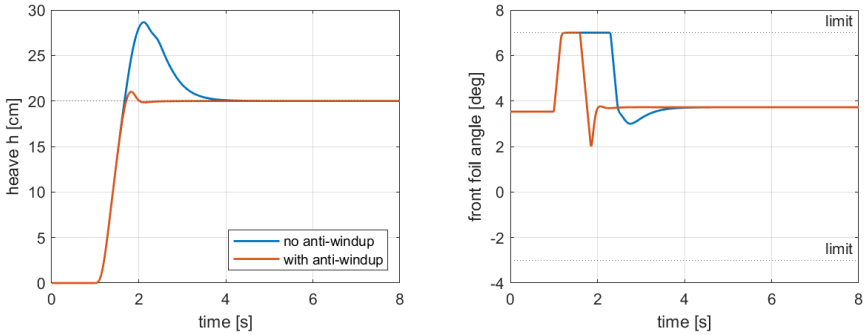


Figure 4: 20 cm step without prefilter: effect of anti-windup.

3 Observer

3.1 Kalman filter with a bias state

The observer is a steady-state Kalman filter for the model augmented with one extra state b, an unknown constant vertical acceleration:

$$\boldsymbol{x}_e = \begin{bmatrix} \boldsymbol{x} \\ b \end{bmatrix}, \qquad \dot{\boldsymbol{x}}_e = A_e \boldsymbol{x}_e + B_e \tilde{\boldsymbol{u}} + L\,(\boldsymbol{y} - C_e \boldsymbol{x}_e), \qquad A_e = \begin{bmatrix} A & \boldsymbol{e}_2 \\ 0 & 0 \end{bmatrix},\ B_e = \begin{bmatrix} B \\ 0 \end{bmatrix},\ C_e = \begin{bmatrix} C & 0 \end{bmatrix}, \tag{3}$$

with $\boldsymbol{e}_2 = [0\ 1\ 0\ 0]^\top$. The input $\tilde{\boldsymbol{u}}$ is the *applied* foil angle, read from the actuator encoders, so the observer stays correct when the actuators saturate. The bias state represents what speed and mass errors do to the boat: the front and rear lift change in the same proportion, which is a pure vertical force with no pitching moment. The augmented pair (A_e, C_e) is observable (rank 5).

3.2 Noise model

- **Process noise**: the wave velocities w_f, w_r enter through E, with variance $(0.11)^2/2$ and an assumed correlation time of 0.1 s; the bias follows a random walk of intensity b_{rate}^2 with $b_{\text{rate}} = 0.5\,\text{m/s}^2/\sqrt{\text{s}}$ (D5).
- **Measurement noise**: for the AHRS and the gyro, sampled noise of variance σ^2 at $T_s = 0.02\,\text{s}$ gives an intensity $\sigma^2 T_s$. The ultrasonic reading also contains the wave surface under the bow, of variance $\zeta_a^2/2$. Its power is concentrated in a band of roughly $\Delta f = 0.25\,\text{Hz}$ around the encounter frequency, so its spectral density there is $\zeta_a^2/(4\Delta f)$, about 100 times the value obtained by treating it as white noise over the sensor bandwidth. We use this larger value (D4).

The resulting observer poles are -0.66, $-0.67 \pm 0.62\text{j}$, -12.0 and -287. The fast poles belong to the pitch states, which are measured directly; the slow ones belong to the heave and the bias, which are inferred mainly from the wave-contaminated height reading.

3.3 Separation principle

With the observer in the loop, the closed-loop poles of the linear system (plant, observer and integrators, 11 states) were computed directly; they equal the LQR poles together with the observer poles to within 5×10^{-13}, as the separation principle predicts. The controller and the observer can therefore be designed independently, and the closed loop is stable.

3.4 Results with output feedback

All wave results in this and the following sections use the corrected wave model of D12.

Estimation (Figure 5). In regular 5 cm waves with sensor noise, after the initial transient the rms estimation errors are 0.72 cm in heave, 1.7 cm/s in heave rate, 0.045° in pitch and 0.23 °/s in pitch rate. The heave estimate follows the true heave and ignores most of the much larger oscillation of the bow-height reading, which is mostly the water surface.

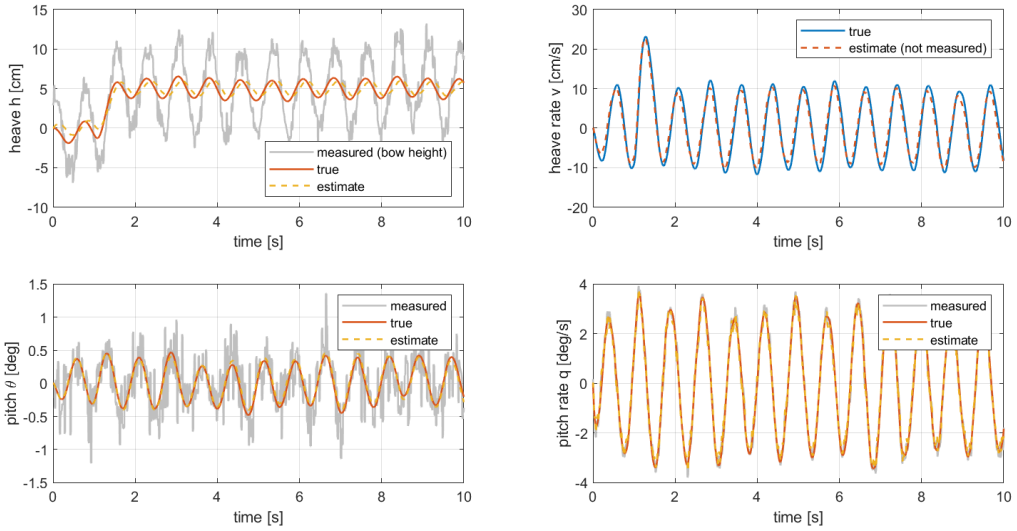


Figure 5: True, estimated and measured states during a 5 cm step in waves with sensor noise. The heave rate is not measured.

Output vs full-state feedback (Figure 6). Under the same step, waves and noise, the height error after settling is 0.95 cm rms with the observer against 0.18 cm with the true state, and the pitch error 0.26° against 0.20°. The difference is the price of not measuring the heave: the observer cannot fully separate the boat's motion from the water surface under the bow. The foils stay well inside their range (front foil peak 1.9°).

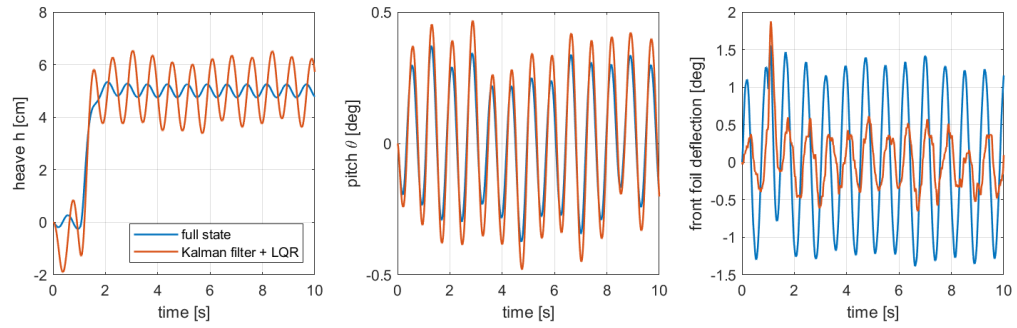


Figure 6: Kalman filter + LQR compared with full-state LQR, 5 cm step in waves with sensor noise.

Robustness (Figure 7). The output-feedback loop is stable over the same speed and mass grid; its slowest pole is −0.57. In nonlinear simulation the height and pitch converge to their targets for speeds of 6 and 8 m/s and masses of 195 and 235 kg (specification S3); after 12 s the largest remaining height error is 0.3 mm. The transient is larger than with full-state feedback, because the bias estimate needs several seconds to build up (D5). The right panel shows the bias state converging to the force caused by the speed error.

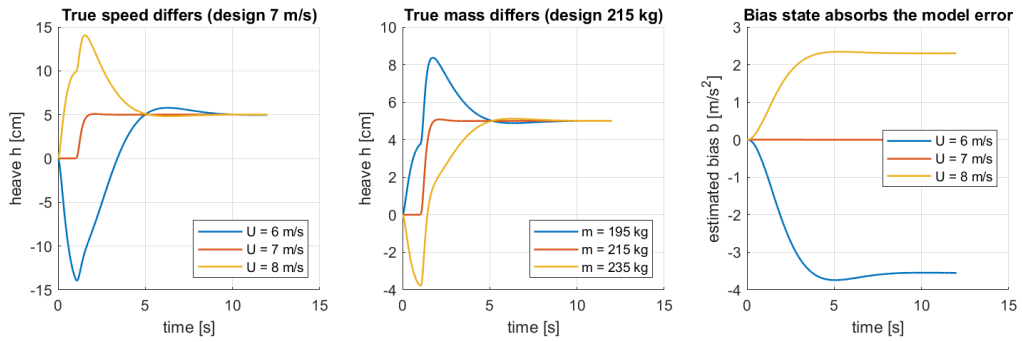


Figure 7: Output feedback with speed and mass errors, and the estimated bias.

4 Digital implementation

The real controller runs on a microcontroller: it reads the sensors at discrete instants, computes, and sends a command that the actuators hold until the next update. This section moves the design to a sampling rate of 50 Hz ($T_s = 20$ ms), the rate assumed for the sensors.

4.1 Why the continuous design cannot simply be sampled

The simplest route, "emulation", runs the continuous controller and observer equations with a numerical integration step of T_s. This fails here. The continuous Kalman filter has a pole at -287 rad/s, associated with the pitch-rate gyro, which is faster than the Nyquist frequency $\pi/T_s = 157$ rad/s. Integrated with forward Euler at 50 Hz, the observer has a discrete pole of magnitude 4.74 and diverges. The observer therefore has to be designed for the sampled system directly.

4.2 Direct digital design

The plant is discretised with a zero-order hold, $\boldsymbol{x}_{k+1} = A_d\boldsymbol{x}_k + B_d\tilde{\boldsymbol{u}}_k$, and both designs are repeated in discrete time with the same weights and noise intensities as before:

- **Discrete LQR** (`dlqr`) on $[\boldsymbol{x};\boldsymbol{\xi}]$, with the integrators updated as $\boldsymbol{\xi}_{k+1} = \boldsymbol{\xi}_k + T_s[\hat{h}_k - r_{f,k};\ y_{2,k}]$ and cost weights QT_s, RT_s.
- **Discrete Kalman filter** (`dlqe`) in current-estimator form. The process-noise covariance is the exact discretisation of the continuous intensity (Van Loan's method), and the measurement covariance is the continuous intensity divided by T_s.

Each sample, the controller performs five steps:

$$
\begin{aligned}
\hat{\boldsymbol{x}}_{k|k-1} &= A_{de}\hat{\boldsymbol{x}}_{k-1|k-1} + B_{de}\big(\bar{\boldsymbol{\delta}}_k - \boldsymbol{u}_0\big) && \text{predict}\\
\hat{\boldsymbol{x}}_{k|k} &= \hat{\boldsymbol{x}}_{k|k-1} + M\big(\boldsymbol{y}_k - C_e\hat{\boldsymbol{x}}_{k|k-1}\big) && \text{correct}\\
\tilde{\boldsymbol{u}}_k &= \boldsymbol{N}_u r_{f,k} - K_d(\hat{\boldsymbol{x}}_{k|k} - \boldsymbol{N}_x r_{f,k}) - K_{i,d}\boldsymbol{\xi}_k - K_u\tilde{\boldsymbol{u}}_{k-1} && \text{control law}\\
\boldsymbol{\xi}_{k+1} &= \boldsymbol{\xi}_k + T_s\big[\hat{h}_{k|k} - r_{f,k};\ y_{2,k}\big] && \text{integrate}\\
r_{f,k+1} &= a\,r_{f,k} + (1-a)\,r_k, \quad a = e^{-T_s/t_f} && \text{prefilter}
\end{aligned}
$$

Here $\bar{\boldsymbol{\delta}}_k$ is the mean of the encoder readings at the start and end of the last interval, the integrators are frozen while the command is saturated, and the term $K_u\tilde{\boldsymbol{u}}_{k-1}$ appears only in the delay-compensated design of Section 4.3. The digital prefilter uses $t_f = 0.6$ s (D10). The equivalent continuous poles of the digital LQR, $\ln z/T_s$, match the continuous design to within 1 % (-3.32, -3.90 ± 2.35j, -10.8 ± 5.6j, -18.4). For the Kalman filter the very fast pitch-rate pole moves from -287 to -178 rad/s (a discrete pole at $z = 0.03$), the complex pair and the pole at -12 are

unchanged, and the slowest real pole moves from -0.66 to -1.28. The digital feedback gains are 10–15 % smaller than the continuous ones.

4.3 Computation delay

A microcontroller needs time to compute, so the command calculated from the measurements at t_k is usually sent at t_{k+1}. We compared three cases on the nonlinear model with sampled sensors and a zero-order-hold output: no delay, a one-sample delay that the design ignores, and a one-sample delay included in the design model as an extra state $\tilde{\boldsymbol{u}}_{k-1}$ (Table 2).

Table 2: Continuous and 50 Hz digital controllers: 5 cm step in 5 cm regular waves with sensor noise (errors after 3 s), and the calm-water step.

Controller	h error (cm rms)	θ (deg rms)	front foil peak (deg)	calm-water step settling (s)	overshoot
Continuous ($t_f = 0.3$ s)	0.95	0.26	1.87	0.72	1.6 %
Digital, no delay	0.97	0.28	1.25	2.21	0
Digital, delay not compensated	1.00	0.32	1.37	2.22	0
Digital, delay compensated (chosen)	0.99	0.33	1.26	2.16	0

The digital controller performs almost as well as the continuous one, and a one-sample delay costs very little: 20 ms is short compared with the closed-loop time constants. We keep the delay-compensated design because its design model matches the implementation (D9). The slower step response of the digital controllers comes from their slower prefilter (D10), not from sampling.

Figure 8 shows the result. The middle panel zooms into half a second: the command changes every 20 ms and the servo follows it with its lag and rate limit. The right panel shows the 50 Hz bow-height samples, dominated by the water surface, and the heave estimate updated at the same rate.

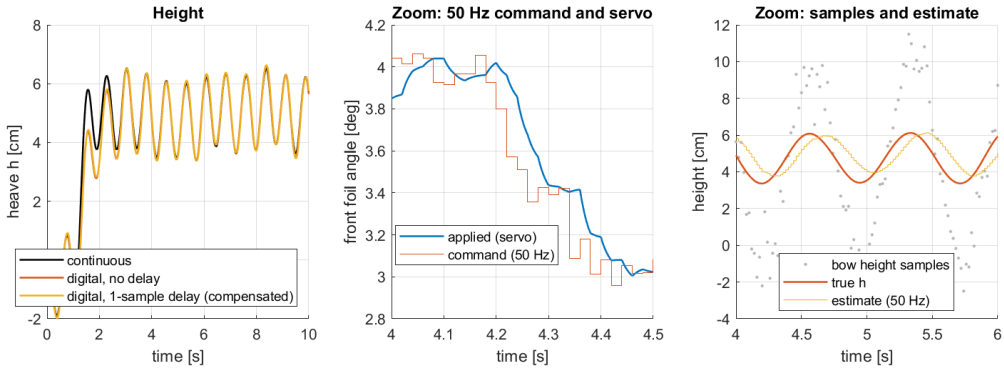


Figure 8: Continuous and 50 Hz digital controllers in waves with sensor noise. Middle: 50 Hz command and servo response. Right: sensor samples, true heave and estimate.

4.4 Sampling rate

Table 3 and Figure 9 repeat the test with the delay-compensated design re-done for each rate (sensors sampled at the same rate, one-sample delay). Performance degrades gradually from 100 Hz down to 15 Hz, where the height error is about 35 % larger than at 50 Hz. At 10 Hz the loop, although stable in the linear analysis, drifts by about 17 cm on the nonlinear model in waves.

Table 3: Effect of the controller rate (5 cm step in regular waves with noise, one-sample delay, delay-compensated design for each rate).

Rate (Hz)	h error (cm rms)	θ (deg rms)	front foil peak (deg)	largest $\lvert z \rvert$
100	0.94	0.30	1.19	0.993
50	0.99	0.33	1.26	0.987
25	1.09	0.39	1.87	0.973
20	1.17	0.45	1.49	0.967
15	1.34	0.49	2.04	0.956
10	16.8	2.42	3.38	0.935

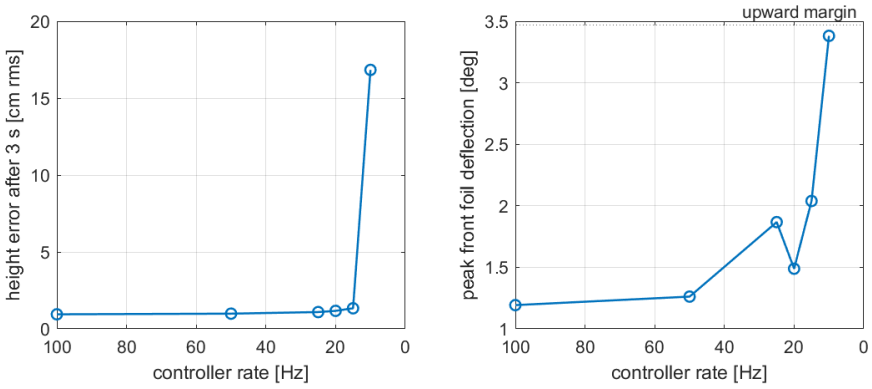


Figure 9: Height error and peak front-foil deflection against controller rate.

4.5 Robustness of the digital loop

The linear closed loop of the sampled plant, digital controller and one-sample delay (13 states) was checked over the same grid of speeds and masses as before; the largest pole magnitude is 0.989, so the loop is stable everywhere. In nonlinear simulation, speeds of 6 and 8 m/s and masses of 195 and 235 kg again give zero steady-state error in height and pitch (Figure 10).

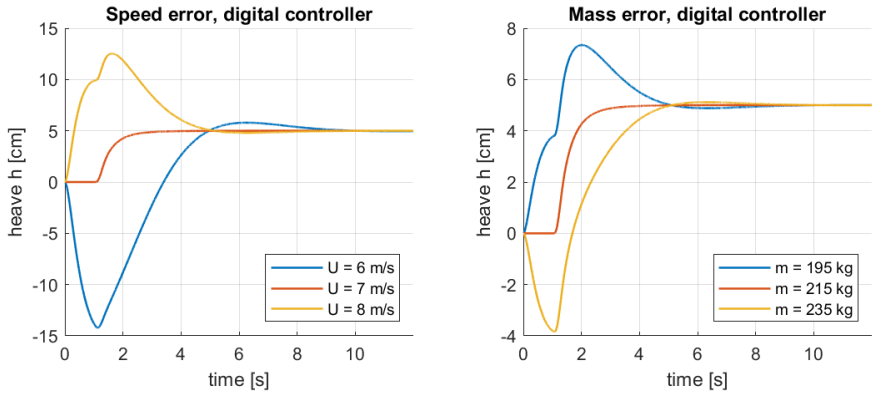


Figure 10: 50 Hz digital controller with speed and mass errors.

4.6 Simulink implementation

In `hydrofoil_sim.slx` the controller is an atomic subsystem with sample time 20 ms. Its inputs are sampled at each tick and its output is held in between, which is the zero-order hold of the real hardware. The algorithm is in a MATLAB Function block that calls `digital_controller.m`: the five steps above, written as they would be on the microcontroller. The encoder input passes

through a memory block, so the controller reads the foil angle from before its own new command; without it, the ideal-actuator option created an algebraic loop. In waves without noise the Simulink model and the MATLAB simulation agree to within 0.34 mm in heave; the difference comes from the encoder reading being one solver step old. With noise the height error is 1.00 cm rms and the pitch 0.33° rms (Figure 11).

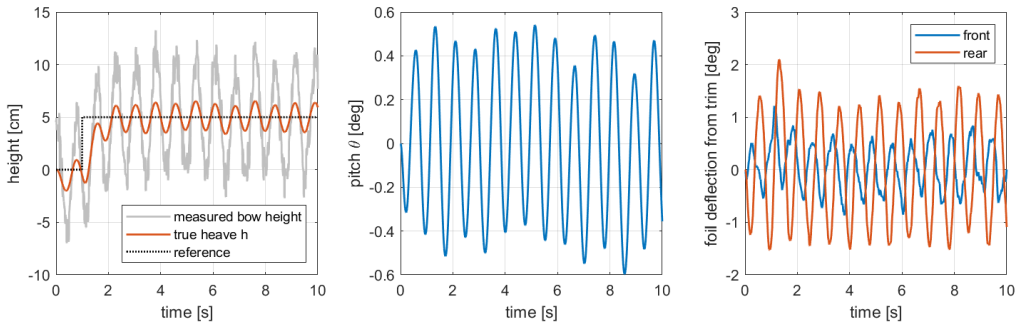


Figure 11: Simulink model with the 50 Hz digital controller: 5 cm step in regular waves with sensor noise.

5 Irregular seas

Regular waves have a single frequency, which the noise model of D4 was built around. Real water surfaces contain a spread of frequencies. We therefore tested the 50 Hz controller in irregular head seas generated from the JONSWAP spectrum (peak-enhancement factor 3.3) as a sum of 80 components with random phases. Three sea states were chosen for the sheltered waters in which a solar boat of this size operates (Figure 12): S1, comparable to the regular wave used so far; S2, moderate; and S3, rough for a boat of this size. Each sea state was run with five different random seas and noise sequences, 40 s each, holding the trim height.

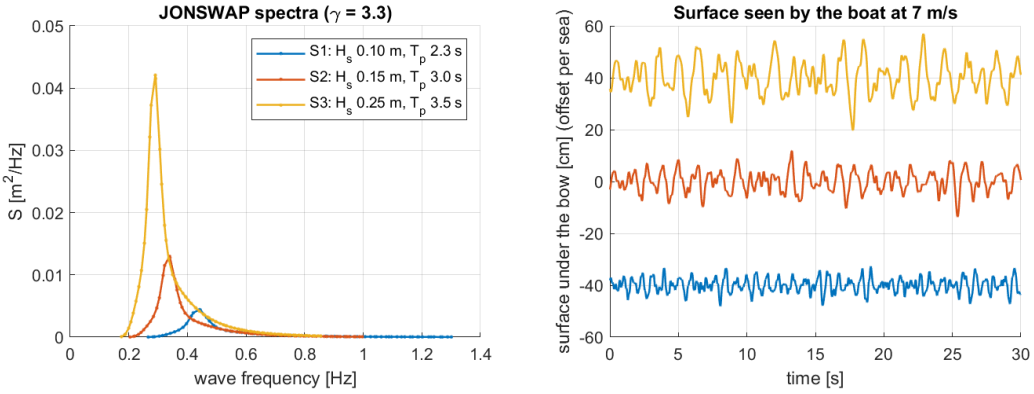


Figure 12: JONSWAP spectra of the three sea states and the surface seen under the bow at 7 m/s.

The results (Table 4, Figure 13) show three things. First, in the gentle sea S1 the performance matches the regular-wave tests, so the narrow-band noise model of D4 still works when the waves are irregular. Second, the height error grows faster than the wave height: from S1 to S3 the significant height increases 2.5 times but the height error about 5 times. The longer waves of S2 and S3 are encountered at lower frequencies, closer to the bandwidth of the heave estimate, so the observer separates them less well from the boat's motion (the estimation error grows from 0.6 to 3–4 cm). This is the same trade-off as D5. Third, even in S3 the largest height deviation, 13 cm, stays inside the 15 cm range of the linear model, and the front foil reaches its limit for

Table 4: 50 Hz controller in irregular head seas: range over five random seas per sea state (statistics after the first 5 s).

Sea	H_s (m)	T_p (s)	h error rms / max (cm)	θ rms (deg)	foil peak (deg)	saturated
S1	0.10	2.3	0.72–0.81 / 2.0–2.8	0.20–0.21	1.5–2.3	0
S2	0.15	3.0	1.8–2.2 / 5.1–5.8	0.25–0.28	1.8–2.5	0
S3	0.25	3.5	3.6–4.9 / 10.5–13.0	0.34–0.40	2.6–4.7	0–0.5 %

at most 0.5 % of the time. The pitch is held within 0.4° rms in all cases, because it is measured directly.

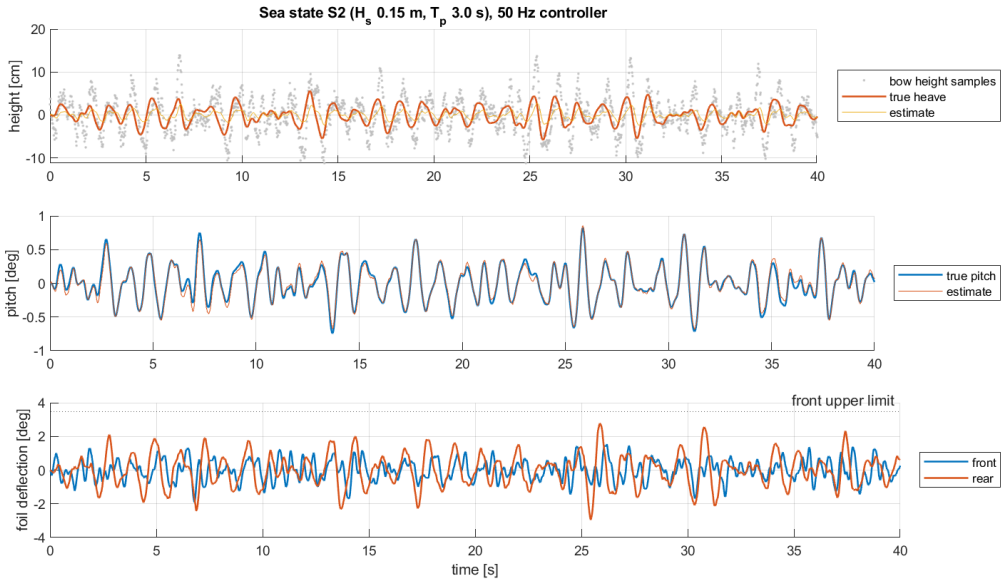


Figure 13: 50 Hz controller in sea state S2: bow-height samples, true and estimated heave and pitch, and foil deflections.

6 Design decisions and trade-offs

The final design was not the first one. Each change below was made because a simulation showed that the earlier choice failed a specification, or, in D12, because an error was found. We record the problem, the evidence, the alternatives considered and the cost of the chosen option. Results involving waves were all recomputed after the correction of D12.

6.1 D1. Reference prefilter

Problem. With a step command, the feedforward and the proportional terms act instantly. **Evidence.** Without the prefilter the 5 cm step has 20.7 % overshoot and asks for 8.9° of front-foil deflection, far beyond the 3.5° margin. **Decision.** A first-order prefilter with $t_f = 0.3$ s in the continuous design: overshoot 1.9 %, peak deflection 1.3°, settling time 0.72 s. **Cost.** None in disturbance rejection, since the prefilter acts only on the reference. The alternative, reducing all gains until a raw step fits the margin, would have slowed the rejection of waves as well.

6.2 D2. Sensor set: two sensors to three

First choice. Bow height and pitch rate. This was based on estimation accuracy only (Table 6): the pitch rate improved the pitch estimates three-fold, and the pitch angle added little more.

Problem. The two-sensor loop did not meet S3. **Evidence.** Table 5 compares observer and integrator options in nonlinear closed loop (calm water) with the speed 1 m/s away from the design value. Both two-sensor variants (A and B) left the boat about 12 cm off in height and 2.4° in pitch, while the observer estimates read zero. With only the height and pitch-rate sensors, a constant force is hard to tell apart from a pitch offset, since both produce the same steady bow-height reading. A single force-bias state is formally observable with these two sensors, but only weakly: the slowest observer pole moved to about -0.005, so the bias would take minutes to converge. A model with separate force and moment biases was not observable at all (rank 5 of 6, with a transmission zero at $s = 0$).

Decision. Add the pitch angle from the AHRS, which the real boat carries. With three sensors, a force bias state and integrators on height and pitch (options F and final), both steady-state errors vanish and the estimates are unbiased. **Cost.** The observer has less left to estimate, but heave and heave rate are still unmeasured, so $n_y = 3 < n_x = 4$ and an observer remains necessary.

Table 5: Steady state in nonlinear closed loop (calm water) with the speed 1 m/s below and above the design value, for different sensor and integrator choices (same LQR weights; observer with the initial noise model).

	Sensors	Observer	Integrators on	$U = 6$ m/s: h, θ	$U = 8$ m/s: h, θ
A	height, rate	KF	bow height (measured)	−12.6 cm, 2.41°	8.0 cm, −1.53°
B	height, rate	KF	$\hat{h}$	−12.6 cm, 2.49°	8.0 cm, −1.53°
C	all three	KF + bias	$\hat{h}$	0, 0.53°	0, −0.33°
D	all three	KF	bow height	−11.4 cm, 2.17°	7.2 cm, −1.38°
E	all three	KF	bow height, θ	0, 0 (but $\hat{h}$ off by 8.9 cm)	0, 0
F	all three	KF + bias	bow height, θ	0, 0	0, 0
Final	all three	KF + bias	$\hat{h}$, θ	0, 0	0, 0

Table 6: Estimation accuracy of a Kalman filter for different sensor sets (one standard deviation; from *model_derivation.pdf*).

Sensors	n_y	h (cm)	v (cm/s)	θ (deg)	q (deg/s)
Height only	1	0.92	5.2	0.22	1.48
Height + pitch rate	2	0.90	4.3	0.06	0.47
All three	3	0.90	4.3	0.03	0.47

Lesson. Rank and estimation variance were not enough to choose the sensors. The decisive test was the steady state of the closed loop under a plausible model error, which is what the integral action is supposed to fix. These results are for calm water and are not affected by D12.

6.3 D3. What the height integrator integrates

First choice. Option F: integrate the measured bow height, the usual recommendation, because an integrator on a measured signal removes the error in that signal whatever the model error. **Problem.** In waves the bow-height reading contains the water surface, and the integrator turns part of it into foil motion: the controller partly follows the surface. **Evidence.** With full-state feedback in regular 5 cm waves, integrating the bow-height reading instead of the true heave raised the height error from 0.18 to 0.60 cm rms and the pitch error from 0.20 to 0.31° rms.

Decision. Integrate the estimated heave $\hat{h}$ instead (with the measured pitch for the second integrator). This is acceptable only because the observer has the bias state, which keeps $\hat{h}$ unbiased under speed and mass errors (option C vs the final row of Table 5). **Cost.** The steady-state accuracy of the height now depends on the observer model: it is exact for force-type errors, but an error the bias state cannot represent would leave a small offset.

6.4 D4. Measurement-noise model for the wave surface

Problem. With the wave surface modelled as white noise of the same variance, the Kalman filter passed much of the surface oscillation into $\hat{h}$ and the loop chased the water. **Evidence.** Table 7: height errors of 3.4–6.8 cm and front-foil peaks up to 6°, beyond the margin.

Alternatives. (a) Set the height-noise intensity to the actual spectral density of the surface signal near the encounter frequency, i.e. assume a bandwidth Δf much smaller than the 25 Hz sensor bandwidth. (b) Add a wave-surface oscillator tuned to the encounter frequency to the observer, a common "wave filter" in ship control. Option (b) was tried with the earlier wave model and performed worse than (a) in the regular wave; it was not repeated after D12, and it would need the encounter frequency, which varies with wavelength and speed.

Table 7: 50 Hz digital controller ($b_{\text{rate}} = 0.5$, $t_f = 0.6$ s): height error (cm rms) and front-foil peak (deg) for different assumed bandwidths of the wave-surface signal. $\Delta f = 25$ Hz is the same as white noise at 50 Hz.

Δf (intensity factor)	regular 8 m	regular 20 m	irregular S1	irregular S2
25 Hz (×1, white)	4.98 / 6.0	5.77 / 3.2	3.41 / 5.1	6.81 / 5.7
2.5 Hz (×10)	1.58 / 2.0	4.09 / 2.4	1.30 / 2.4	3.32 / 3.2
0.25 Hz (×100, chosen)	0.98 / 1.5	2.69 / 1.7	0.81 / 1.4	2.11 / 1.7
0.08 Hz (×310)	0.93 / 1.5	2.27 / 1.5	0.74 / 1.2	1.82 / 1.4

Decision. Option (a) with $\Delta f = 0.25$ Hz, a physically reasonable bandwidth for the encounter spectrum. A narrower bandwidth helps a little more in long waves, so the choice is not critical. **Cost.** The heave estimate becomes slow (observer poles near -0.66), see D6.

6.5 D5. Speed of the bias state

Trade-off. The bias random-walk intensity sets how fast the observer attributes a persistent height error to a model error. A fast bias recovers quickly from a speed or mass error, but long waves, encountered at low frequency, then also look like a slowly varying bias, and the controller reacts to them.

Table 8: Effect of the bias drift rate b_{rate} (50 Hz digital controller, $\Delta f = 0.25$ Hz, $t_f = 0.6$ s). Speed errors: height deviation during a 5 cm step with the speed 1 m/s off the design value. Waves: height error in cm rms.

	Speed error: peak / back within 5 mm		Waves with noise		
b_{rate}	$U = 6$ m/s	$U = 8$ m/s	regular 8 m	regular 20 m	irregular S2
0.3	21.5 cm / 10.1 s	11.6 cm / 6.2 s	0.72	2.33	1.76
0.5 (chosen)	19.2 cm / 7.6 s	9.9 cm / 4.4 s	0.98	2.69	2.11
1.0	15.9 cm / 5.1 s	7.5 cm / 2.8 s	1.55	3.40	2.82
3.0	10.4 cm / 2.1 s	—	4.85	5.28	6.17

Decision. $b_{\text{rate}} = 0.5$, a compromise. A sudden 1 m/s speed change is a severe test (in practice the speed changes gradually), while waves are always present. **Possible improvement.** The real boat measures its speed by GPS. Scheduling the trim angles $\boldsymbol{u}_0(U)$ on the measured speed would remove most of the speed error before it reaches the observer, and the bias state

could then be made slower still. We did not do this, to keep the speed out of the set of feedback measurements.

6.6 D6. Observer slower than the controller

The usual rule of thumb places the observer poles 2–5 times faster than the controller poles. Here the slowest observer pole (-0.66) is about five times *slower* than the slowest controller pole (-3.3). This is the direct result of D4 and D5: the Kalman filter weighs the wave-contaminated height reading lightly, so the heave estimate is slow. A faster observer would follow the water surface. The separation principle still guarantees stability, and the pitch states, which are measured directly, have fast observer poles (-12 and -287). The cost is that disturbances which appear only in the height, such as a sudden change of speed, are corrected more slowly than the controller alone would allow (Table 8), and that long waves are rejected less well (Section 5).

6.7 D7. Simulink: zero-crossing detection

The actuator position and rate limits switch often, and with the continuous controller the anti-windup switch did too. Simulink's zero-crossing detection located thousands of consecutive events at these switches and stopped. It was disabled and the maximum step kept at 5 ms, which is how the MATLAB simulations treat the same switches.

6.8 D8. Digital observer instead of emulation

Problem. The continuous Kalman filter has a pole at -287 rad/s, beyond the 157 rad/s Nyquist frequency at 50 Hz; run with forward Euler it diverges ($|z| = 4.74$). **Alternatives.** Raise the rate above 100 Hz, implement the observer with an exact matrix exponential, or design it directly in discrete time. **Decision.** Direct digital design of both the LQR and the Kalman filter with the same weights. **Cost.** The design now depends on the sampling rate and must be redone if the rate changes; the functions take T_s as an input for that reason.

6.9 D9. Computation delay

Decision. Assume the command is sent one sample after the measurement and include that delay in the LQR design model. **Evidence.** Table 2: the delay changes the height error by only 0.03 cm, compensated or not, because 20 ms is short compared with the closed-loop time constants. We keep the compensation because the design model then matches the implementation, and because it would matter more at lower rates.

6.10 D10. Prefilter of the digital controller

History. The first digital tests, made with the wave model that D12 later corrected, showed the front foil saturating in waves during a step with $t_f = 0.3$ s, and the prefilter was slowed to 0.6 s. With the corrected wave model that saturation does not occur: with $t_f = 0.3$ s the front foil peaks at 2.6–3.0° over five noise sequences, and with 0.6 s at 1.8–2.1° ($b_{\text{rate}} = 1$, regular 5 cm waves); the height error is the same, 1.5–1.6 cm.

Decision. Keep $t_f = 0.6$ s. It still meets S2 (settling in about 2.2 s, no overshoot) and leaves about 1° more of the 3.5° margin for larger waves; in sea state S3 the foil already reaches its limit for short moments. **Cost.** Step commands are slower than with the continuous design (2.2 s against 0.7 s).

6.11 D11. Sampling rate

Decision. 50 Hz. **Evidence.** Table 3: the height error grows slowly as the rate is reduced and the loop fails at 10 Hz; 50 Hz is five times above that point, and 100 Hz improves the error by

only 0.05 cm. **Cost.** A 50 Hz loop with a 5-state observer and a few small matrix products is a light load for a microcontroller such as the Teensy board used on SR02; the ultrasonic sensor must also deliver readings at 50 Hz, which should be confirmed for the actual sensor.

6.12 D12. Error in the wave model, found and corrected

Problem. While generalising the wave model to irregular seas, we found a sign error in the vertical water velocity. For a wave travelling towards the bow, $\zeta = a\cos(kx + \omega t)$, the vertical water velocity is $\partial\zeta/\partial t \cdot e^{-kd} = -a\omega e^{-kd}\sin(kx + \omega t)$; the first implementation had the opposite sign. This changed the phase between the wave force on the boat and the wave surface seen by the height sensor by 180°.

Effect. Results with full-state feedback, calm water or open loop without sensors were unaffected. Results combining waves with the observer changed substantially: in regular 5 cm waves the output-feedback height error of the digital controller rose from about 0.75 to 1.0 cm rms, while the front-foil peaks fell from about 3.4 to 1.3°. Three earlier conclusions changed as a result: the bias drift rate was reduced from 1 to 0.5 (D5); the saturation that had motivated the slower prefilter disappeared (D10); and the apparent need for 50 Hz because of saturation at 25 Hz disappeared, the rate now being chosen from a gradual trade-off (D11). The measurement-noise model (D4), the sensor choice (D2) and the integrator choice (D3) were confirmed.

Fix and check. All wave calculations now go through one function, `wave_eval.m`, used by the MATLAB simulations and by the Simulink model. It was verified by checking that its water velocity equals the time derivative of its surface elevation times e^{-kd} (difference 0.3 %, from the numerical derivative). **Lesson.** A sign error in a disturbance model cannot be found by comparing two implementations of the same equations, as our Simulink and MATLAB checks did; it needs a physical check of the model itself.

7 Discussion: constraints and uncertainty

- **Actuator margin.** The front foil has only 3.5° of upward travel at 7 m/s, and less at lower speed. In the gentle and moderate sea states the controller uses at most about 2.5° of it; in the rough sea S3 the foil reaches its limit for up to 0.5 % of the time. The prefilter and anti-windup keep the response graceful when this happens (Figure 4), but a wider mechanical range or a higher cruise speed would be the real fix.

- **Unmeasured heave.** Every result with the observer is worse than with the true state, and the gap grows with the wave period (Section 5). With only a bow-height sensor that also sees the water surface, long waves and slow model errors are hard to separate from the boat's own motion. A second height sensor at another position, or an accelerometer, would help most.

- **Model uncertainty.** Stability is robust over the speed and mass range tested, and steady-state errors vanish thanks to the integrators and the bias state. The transient after a sudden speed change is the weakest point (D5).

- **Assumed parameters.** The chord, submergence, noise levels and actuator rate were estimated (see *model_derivation.pdf*). The design choices above depend mostly on the ratio between the wave-surface signal and the boat motion, which is a physical feature of the problem rather than of these values.

8 Summary

The final controller is an LQR with integral action on the estimated height and the measured pitch, a reference prefilter and anti-windup, combined with a Kalman filter that estimates the

four states and a constant force bias from three sensors. It is implemented as a 50 Hz digital controller designed directly in discrete time, including a one-sample computation delay. On the nonlinear model with actuator limits, waves and sensor noise it meets all five specifications: a 5 cm step settles in about 2 s without overshoot, the height is held within about 1 cm rms in 5 cm regular waves and in an irregular sea of 0.10 m significant height, and speed and mass errors leave no steady-state error. The main trade-offs are between rejecting the water surface and reacting quickly to model errors, and between performance and the small upward foil margin.